AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 15 · WORKBOOK

Save New Examples for Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected evaluation record.

Identify data roles, preserve original results, expose repeated-score selection and inspect coverage.

Keep one habit: Keep each model version, prediction record and use of feedback visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Build a Human Classifier

Debug the Cause

Algorithms as Recipes

W1 · Identify the role of each example use

> Separate fitting, frozen evaluation and feedback-guided revision.

Your evidence record

- For each event, state whether examples are being used for development or independent evaluation of a fixed model: comparing candidates on **fitting** labels; scoring a frozen model on previously reserved examples once; changing a threshold after reading a test score.

  Explain your last choice.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Preserve the original model result

> Keep TowerA's prediction record alongside TowerB's seen-case replay.

Your evidence record

- TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its **prediction**s, agreements and disagreement.

  Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Trace test information into selection

> Explain leakage through repeated score comparisons.

Your evidence record

- A team leaves reserved labels out of its **fitting** table but tries six rules and reads a test score for each. It chooses the highest-scoring rule. Explain how information still reached model selection.

  Would renaming or reordering the same test records repair the problem?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Inspect duplication and coverage

> Propose new appropriate cases without inventing a result.

Your evidence record

- A class reserves ten copies of one easy development example in a new file. The model agrees with all ten labels.

  Explain two reasons this gives weak **evidence** about new cases, and propose a better final evaluation plan without promising perfect future performance.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Run a separate ribbon experiment

> Keep one first attempt before hints or feedback.

Read and explain independently

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Run a separate ribbon experiment

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Preserve the original result for Keeping final tests unseen. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Keeping final tests unseen. Proposed explanation: “A new file makes old records unseen”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 15 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

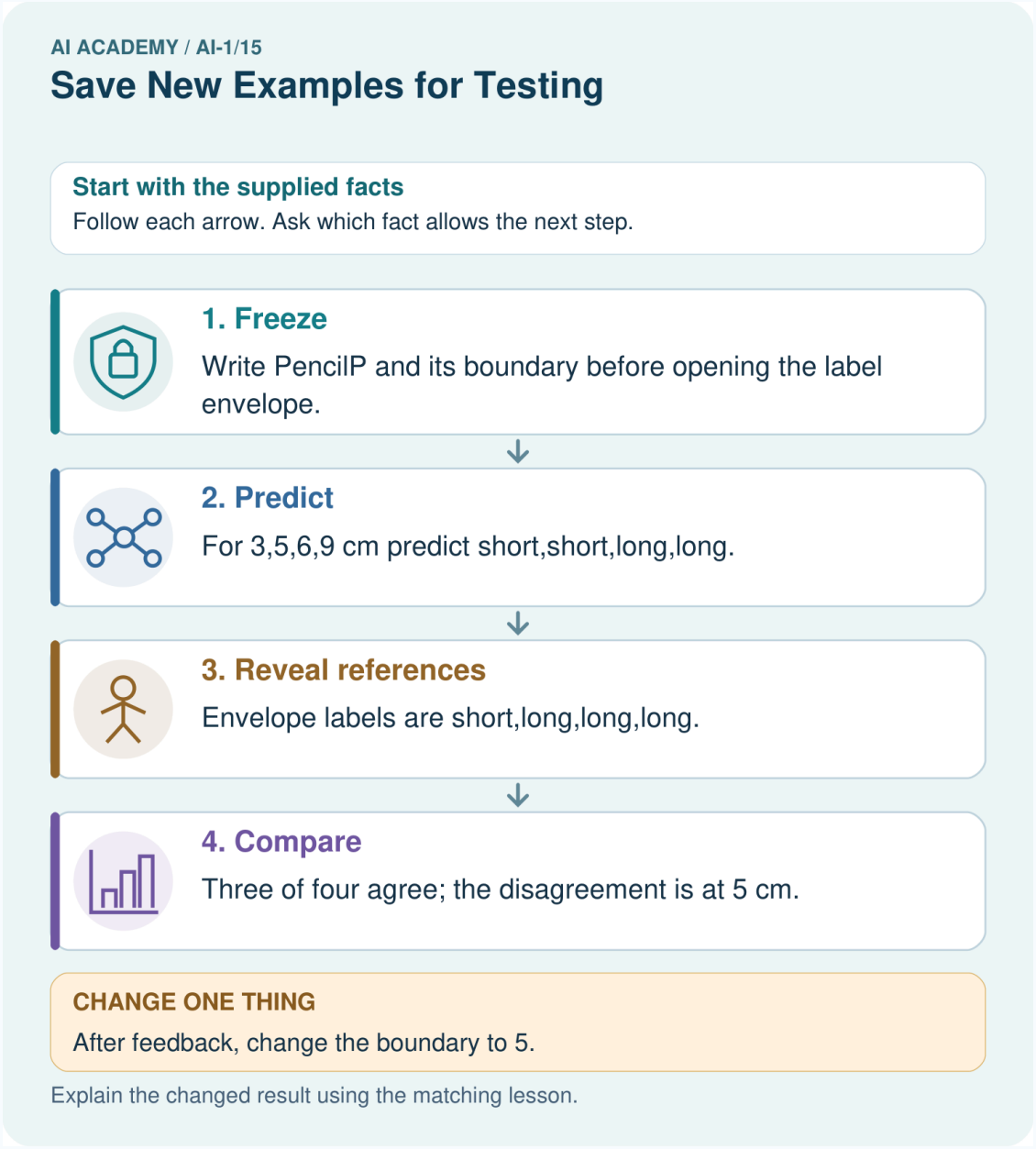
AI ACADEMY / AI-1/15

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Does 4/4 on the replay replace the original 3/4 result? Explain.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional envelope exercise evaluates a fixed pencil-length sorter.
- Which labelled step matches this detail? Three of four agree; the disagreement is at 5 cm.
- 1. Reveal references
- 2. Compare
- 3. **Freeze**
- 4. Predict
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare Three of four agree; the disagreement is at 5 cm. The **prediction** record must exist before the labels are seen.

PencilP scores 3/4=75% agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Changed-case question

- Changed-case practice: After feedback, change the boundary to 5. Which conclusion is supported by the supplied facts?
- 1. The revised rule matches all four known labels but needs a different final test.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The revised rule matches all four known labels but needs a different final test.

These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Repair and save your work

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Save New Examples for Testing: the case

AI ACADEMY / AI-1/15 · 01 · LOOK AT THE FACTS

Save New Examples for Testing: the case

The situation

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Supplied evidence

Freeze PencilP: long at least 6 cm; short below 6.
Reserved valid lengths are 3,5,6,9 cm.
Teacher envelope labels, revealed only after predictions, are short,long,long,long.

By the end, I can freeze a model before labels are revealed and report a bounded test result.

Follow the labelled picture with the matching explanation. Open the vector diagram

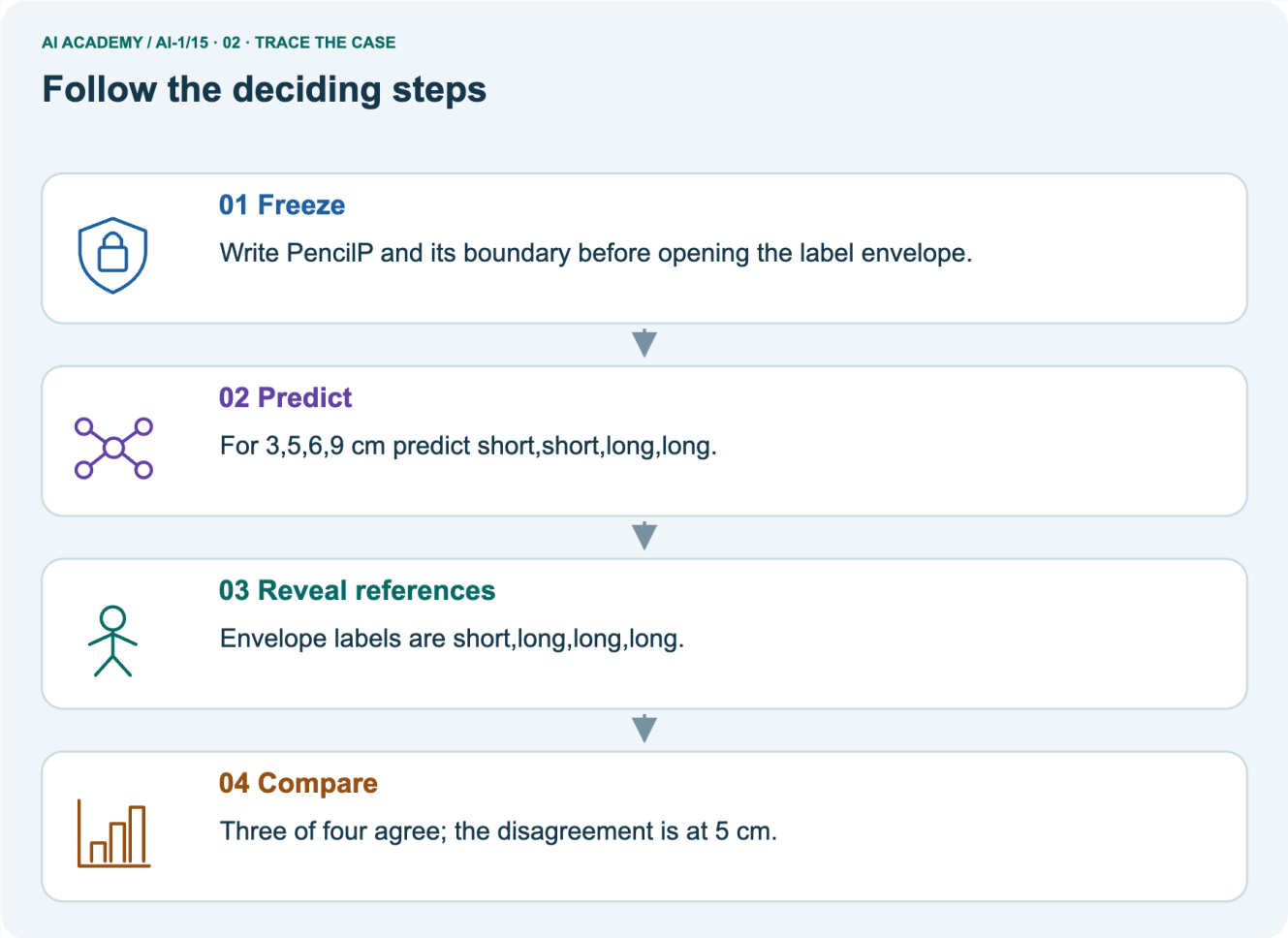
A fictional envelope exercise evaluates a fixed pencil-length sorter. Supplied facts: **Freeze** PencilP: long at least 6 cm; short below 6. Reserved valid lengths are 3,5,6,9 cm.

Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



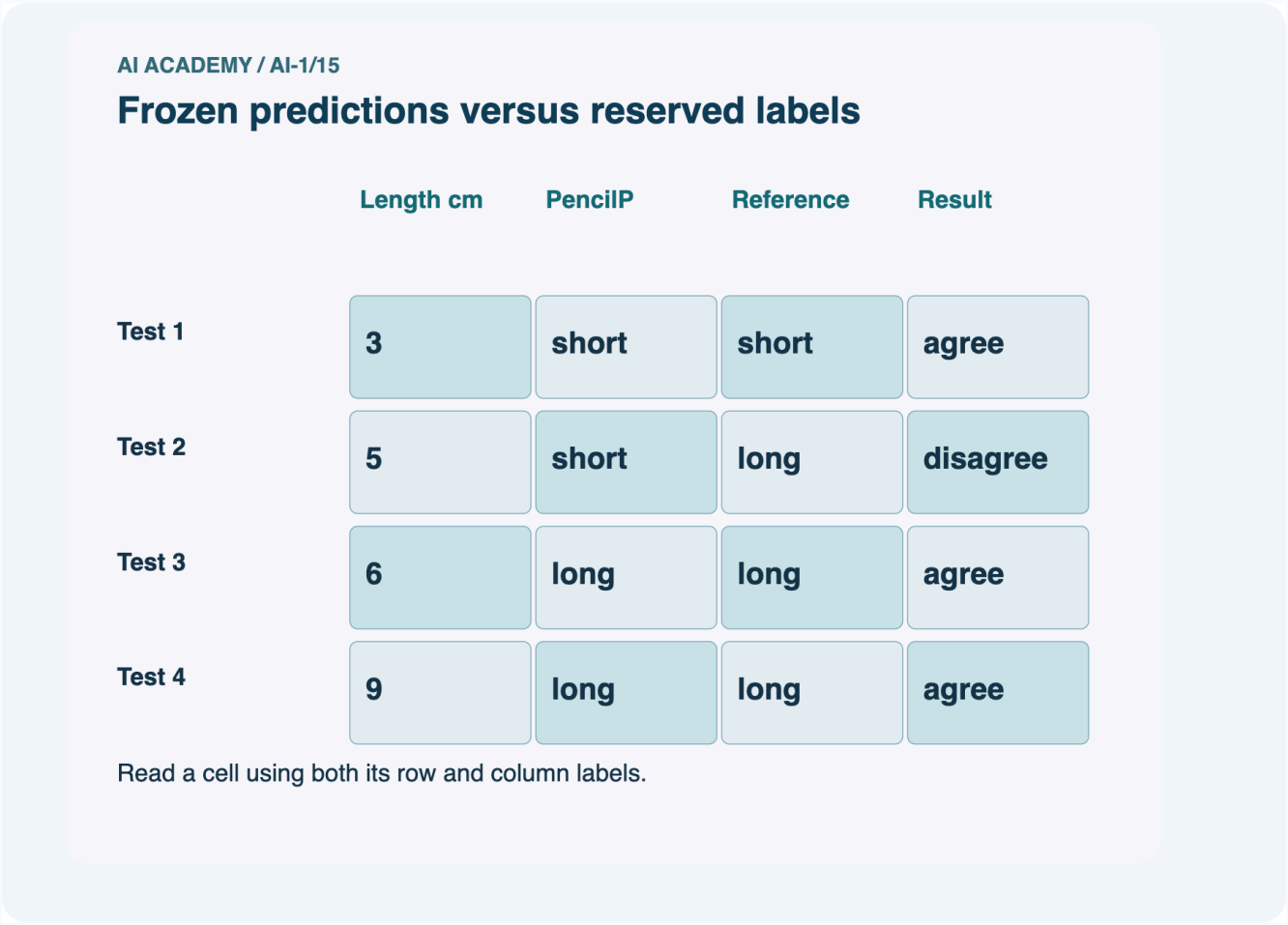
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze: Write PencilP and its boundary before opening the label envelope. → Predict: For 3,5,6,9 cm predict short,short,long,long. → Reveal references: Envelope labels are short,long,long,long. → Compare: Three of four agree; the disagreement is at 5 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Frozen predictions versus reserved labels



Follow the labelled picture with the matching explanation. Open the vector diagram

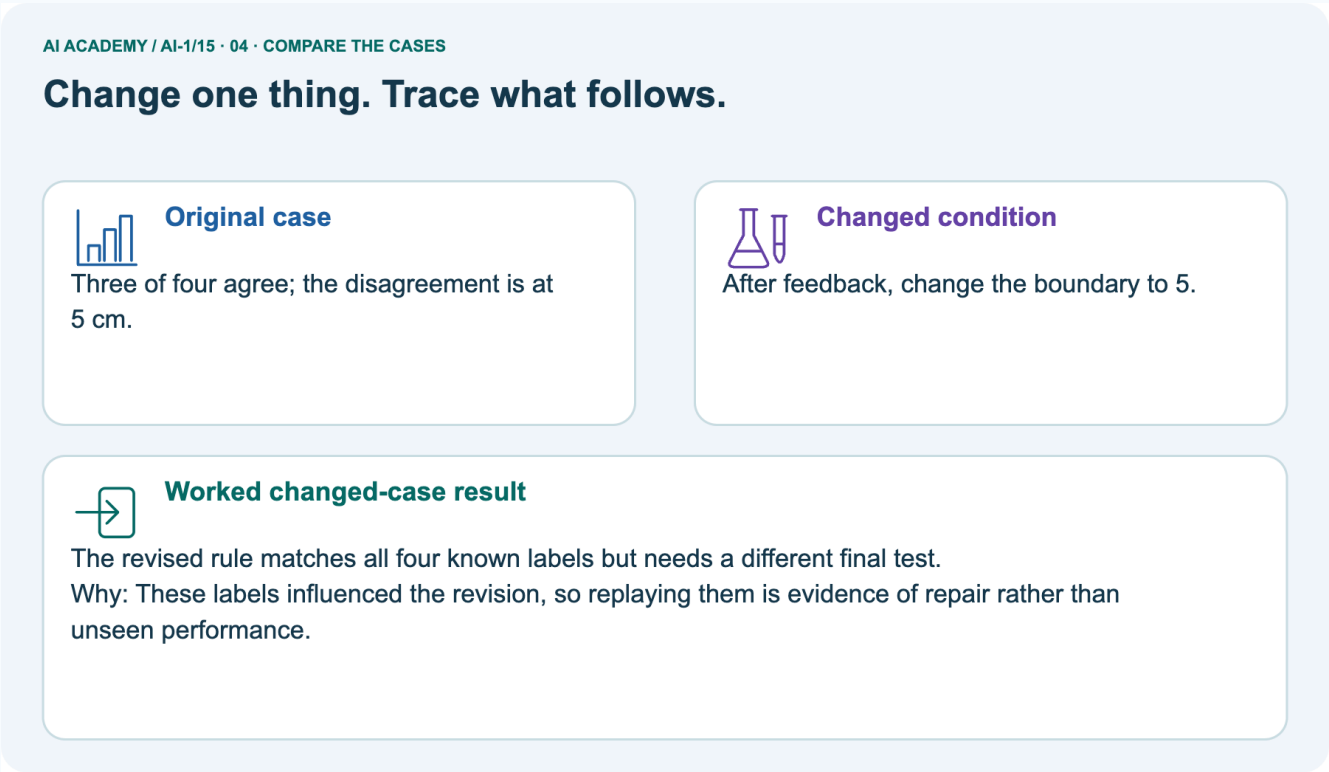
Frozen **prediction**s versus reserved labels. Test 1: Length cm = 3, PencilP = short, Reference = short, Result = agree. Test 2: Length cm = 5, PencilP = short, Reference = long, Result = disagree.

Test 3: Length cm = 6, PencilP = long, Reference = long, Result = agree. Test 4: Length cm = 9, PencilP = long, Reference = long, Result = agree.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

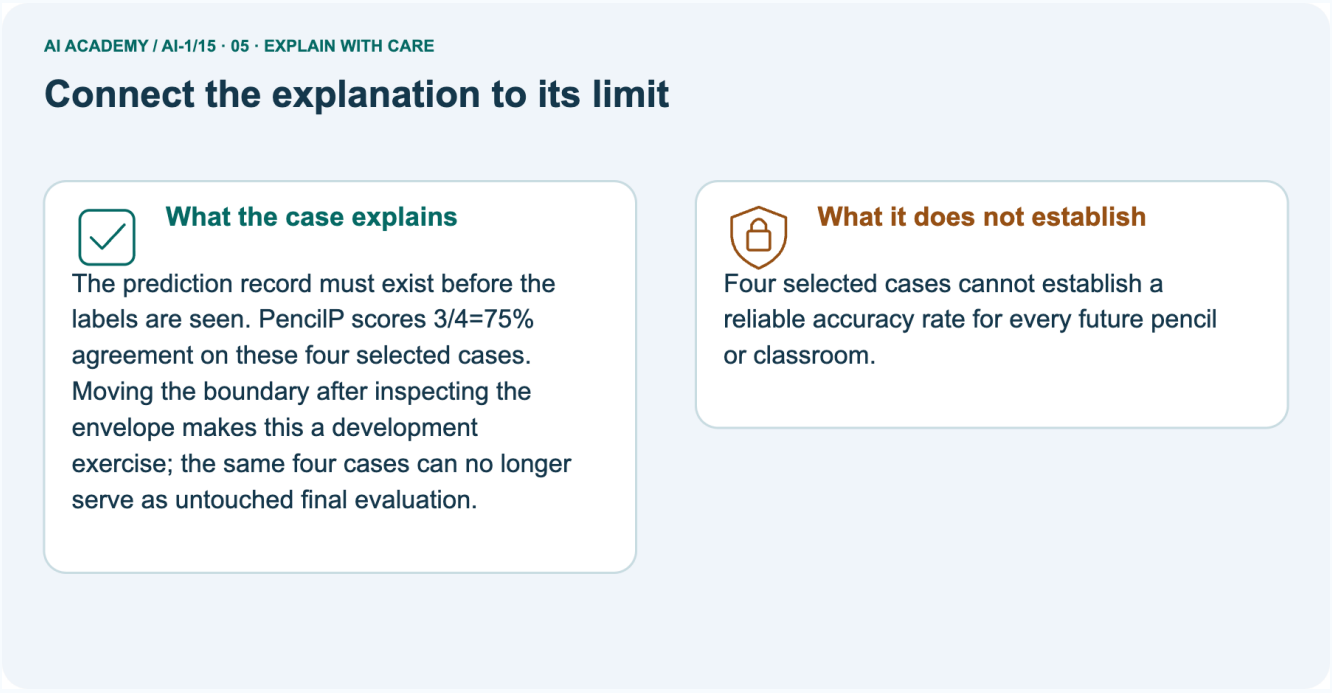
Original case: Three of four agree; the disagreement is at 5 cm. Changed condition: After feedback, change the boundary to 5. Expected result: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

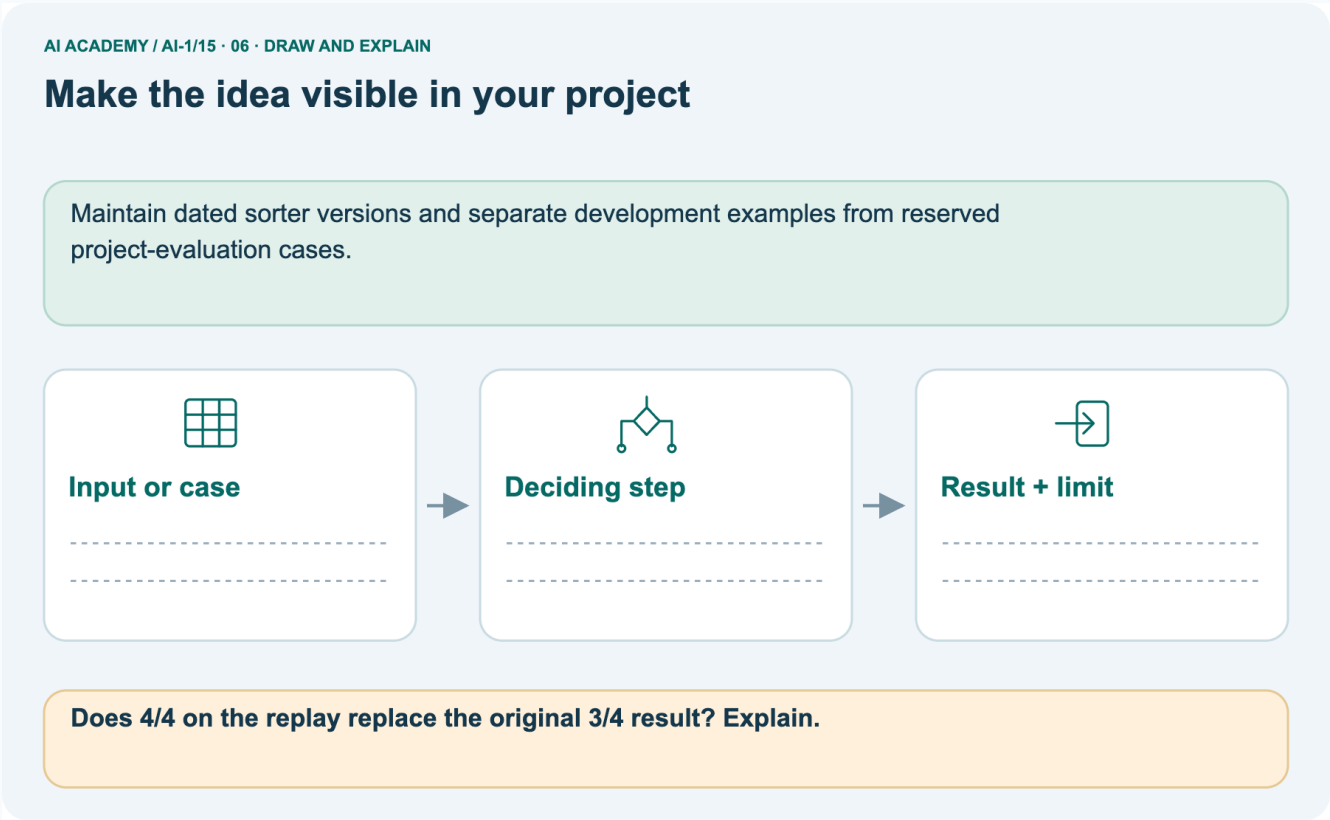
Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Limit: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases. Draw your **input**, deciding step, result and limit.

Supported question: Does 4/4 on the replay replace the original 3/4 result? Explain.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Save New Examples for Testing: the case

02. Follow the deciding steps

03. Frozen predictions versus reserved labels

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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